Math 241, F18
Exam 1

Name: Answer

<u>**Instructions:**</u>

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the exam.

Problem	Points	Score
1	20	
2	24	
3	20	
4	16	
5	20	
Total	100	

Good Luck !

Problem 1. (20 points) Given two vectors $\mathbf{a} = < \sqrt{3},\ 3,\ 2\sqrt{6} >$ and $\mathbf{b} = < -\sqrt{3},\ 0,\ \sqrt{6} >$.
(a) Find $|\mathbf{a}|$ and $|\mathbf{b}|$.

$$|a| = \sqrt{(\sqrt{3})^2 + 3^2 + (2\sqrt{6})^2} = \sqrt{3+9+24} = \sqrt{36} = 6$$

$$|b| = \sqrt{(-\sqrt{3})^2 + 0^2 + (\sqrt{6})^2} = \sqrt{3+0+6} = \sqrt{9} = 3$$

(b) Find a unit vector $\mathbf{u}$ which has the same direction as $\mathbf{a}$.

$$u = \frac{a}{|a|} = \frac{<\sqrt{3}, 3, 2\sqrt{6}>}{6} = <\frac{\sqrt{3}}{6}, \frac{1}{2}, \frac{\sqrt{6}}{3}>$$

(c) Find the angle θ between $\mathbf{a}$ and $\mathbf{b}$.

$$a \cdot b = \sqrt{3} \cdot (-\sqrt{3}) + 3 \cdot 0 + 2\sqrt{6} \cdot \sqrt{6} = -3 + 0 + 12 = 9$$

$$\cos\theta = \frac{a \cdot b}{|a||b|} = \frac{9}{3 \times 6} = \frac{1}{2}$$

$$\Rightarrow \theta = \cos^{-1}\left(\frac{1}{2}\right) = \frac{\pi}{3} \text{ or } 60^\circ$$

(d) Find the vector projection of $\mathbf{b}$ onto $\mathbf{a}$, $\text{proj}_{\mathbf{a}}\mathbf{b}$.

$$\text{Proj}_a b = \left(\frac{a \cdot b}{|a|^2}\right) a$$

$$= \frac{9}{6^2} \cdot <\sqrt{3}, 3, 2\sqrt{6}>$$

$$= <\frac{\sqrt{3}}{4}, \frac{3}{4}, \frac{\sqrt{6}}{2}>$$

Problem 2. (24 points) Given three points $P(-1,3,1)$, $Q(0,5,-2)$, and $R(4,3,-1)$.
(a) Find parametric equations for the line through the points P and Q.

$$\overrightarrow{PQ} = \langle 0,5,-2\rangle - \langle -1,3,1\rangle = \langle 1,2,-3\rangle$$

$$\Rightarrow r(t) = \langle -1,3,1\rangle + t\langle 1,2,-3\rangle$$

$$x(t) = -1+t,\quad y(t) = 3+2t,\quad z(t) = 1-3t,$$
$$-\infty < t < \infty$$

(b) Find an equation for the plane through the point P normal to the vector $2\mathbf{i}-\mathbf{j}+3\mathbf{k}$.

$$\vec{n} = \langle 2,-1,3\rangle$$

$$2(x+1) - 1\cdot(y-3) + 3(z-1) = 0$$

$$2x+2-y+3+3z-3 = 0$$

$$2x - y + 3z = -2$$

(c) Find an equation for the plane that contains the triangle $\triangle PQR$.

$$\overrightarrow{PQ} = \langle 1,2,-3\rangle$$

$$\overrightarrow{PR} = \langle 4,3,-1\rangle - \langle -1,3,1\rangle = \langle 5,0,-2\rangle$$

$$\vec{n} = \overrightarrow{PQ}\times\overrightarrow{PR} = \begin{vmatrix} i & j & k \\ 1 & 2 & -3 \\ 5 & 0 & -2 \end{vmatrix} = i\begin{vmatrix} 2 & -3 \\ 0 & -2 \end{vmatrix} - j\begin{vmatrix} 1 & -3 \\ 5 & -2 \end{vmatrix} + k\begin{vmatrix} 1 & 2 \\ 5 & 0 \end{vmatrix}$$

$$= -4i - 13j - 10k$$

Thus the equation is given by

$$-4(x+1) - 13(y-3) - 10(z-1) = 0$$

$$-4x - 13y - 10z = -45$$

$$4x + 13y + 10z = 45$$

Problem 3. (20 points)
(a) Find the velocity vector $\mathbf{v}(t)$ and the position vectors $\mathbf{r}(t)$ of a particle that has the acceleration $\mathbf{a}(t) =< 2t,\ e^t,\ e^{-t} >$ and initial conditions $\mathbf{v}(0) =< 0, 1, -1 >$ and $\mathbf{r}(0) =< 1, 0, 0 >$.

$$V(t) = \int a(t)dt = \int <2t, e^t, e^{-t}> dt$$
$$= <t^2, e^t, -e^{-t}> + C_1$$
$$V(0) = <0,1,-1> \Rightarrow C_1 = <0,1,-1> - <0,1,-1> = 0$$
$$\Rightarrow V(t) = <t^2, e^t, -e^{-t}>$$
$$r(t) = \int v(t)dt = \int <t^2, e^t, -e^{-t}> dt$$
$$= <\frac{t^3}{3}, e^t, e^{-t}> + C_2$$
$$r(0) = <1,0,0> \Rightarrow C_2 = <1,0,0> - <0,1,1> = <1,-1,-1>$$
$$\Rightarrow r(t) = <\frac{t^3}{3}, e^t, e^{-t}> + <1,-1,-1>$$
$$= <\frac{t^3}{3}+1, e^t-1, e^{-t}-1>$$

(b) Let $\mathbf{r}(t) = 2t\,\mathbf{i} + t^2\,\mathbf{j} - \frac{1}{3}t^3\,\mathbf{k}$, find the arc length of the curve segment defined by $\mathbf{r}(t)$ with $0 \leq t \leq 3$.

$$r'(t) = 2i + 2tj - t^2k$$
$$|r'(t)| = \sqrt{2^2 + (2t)^2 + (-t^2)^2} = \sqrt{4 + 4t^2 + t^4}$$
$$= \sqrt{(2+t^2)^2} = 2 + t^2$$
$$\Rightarrow L = \int_0^3 (2+t^2)dt = \left[2t + \frac{t^3}{3}\right]_{t=0}^{t=3}$$
$$= \left[6 + \frac{27}{3}\right] - 0$$
$$= 15$$

Problem 4. (16 points) Given a space curve $\mathbf{r}(t) =< e^t \cos t,\, e^t \sin t,\, 2 >$.
(a) Find the unit tangent vector $\mathbf{T}$ of the curve.

$$r'(t) = \langle e^t \cos t - e^t \sin t,\, e^t \sin t + e^t \cos t,\, 0 \rangle$$
$$= \langle e^t(\cos t - \sin t),\, e^t(\sin t + \cos t),\, 0 \rangle$$
$$|r'(t)| = \sqrt{e^{2t}(\cos t - \sin t)^2 + e^{2t}(\sin t + \cos t)^2 + 0^2} \leftarrow \cos^2 t + \sin^2 t = 1$$
$$= \sqrt{e^{2t}(1 - 2\cos t \sin t) + e^{2t}(1 + 2\sin t \cos t)}$$
$$= \sqrt{2e^{2t}} = \sqrt{2}e^t$$
$$\Rightarrow T(t) = \frac{r'(t)}{|r'(t)|} = \frac{\langle e^t(\cos t - \sin t),\, e^t(\sin t + \cos t),\, 0\rangle}{\sqrt{2}e^t}$$
$$= \frac{1}{\sqrt{2}}\langle \cos t - \sin t,\, \sin t + \cos t,\, 0 \rangle$$

(b) Find the curvature κ and the principal unit normal vector $\mathbf{N}$ of the curve.

$$T'(t) = \frac{1}{\sqrt{2}}\langle -\sin t - \cos t,\, \cos t - \sin t,\, 0 \rangle$$
$$|T'(t)| = \sqrt{\tfrac{1}{2}(-\sin t - \cos t)^2 + \tfrac{1}{2}(\cos t - \sin t)^2 + 0^2}$$
$$= \sqrt{\tfrac{1}{2}(1 + 2\sin t \cos t) + \tfrac{1}{2}(1 - 2\cos t \sin t)}$$
$$= \sqrt{1} = 1$$
$$\Rightarrow k(t) = \frac{|T'(t)|}{|r'(t)|} = \frac{1}{\sqrt{2}e^t} = \frac{e^{-t}}{\sqrt{2}}$$
$$N(t) = \frac{T'(t)}{|T'(t)|} = \frac{1}{\sqrt{2}}\langle -\sin t - \cos t,\, \cos t - \sin t,\, 0 \rangle$$

Problem 5. (20 points) Find the limit if it exists or show the limit does not exist.

(a) $\lim_{(x,y)\to(0,1/6)} e^{-x}\sin(\pi y)$

$= e^{-0}\sin(\frac{\pi}{6})$

$= 1\cdot\frac{1}{2} = \frac{1}{2}$

(b) $\lim_{(x,y)\to(1,0)} \ln\left(\frac{1+y^2}{x^2+xy}\right)$

$= \ln\left(\frac{1+0^2}{1^2+1\cdot 0}\right) = \ln 1 = 0$

(c) $\lim_{(x,y)\to(0,0)} \frac{x^2y}{2x^3+y^3}$

Consider the values along the line $y=kx$

$\frac{x^2y}{2x^3+y^3} = \frac{x^2(kx)}{2x^3+(kx)^3} = \frac{kx^3}{2x^3+k^3x^3} = \frac{k}{2+k^3}$

When $k=1$, $\frac{k}{2+k^3} = \frac{1}{2+1^3} = \frac{1}{3}$

when $k=-1$, $\frac{k}{2+k^3} = \frac{-1}{2+(-1)^3} = -1$

$\Rightarrow$ the limit doesn't exist.

(d) $\lim_{(x,y)\to(1,1)} \frac{x^2-y^2}{x-y}$

$= \lim_{(x,y)\to(1,1)} \frac{\cancel{(x-y)}(x+y)}{\cancel{x-y}} = \lim_{(x,y)\to(1,1)} x+y = 1+1 = 2$